

# Project 1: Design of a Heat Exchanger

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## 1 Introduction and Process Description

The Ilam Gas Refinery is responsible for treating sour gas and separating impurities such as  $\text{CO}_2$  and  $\text{H}_2\text{S}$ . In the gas sweetening unit, in order to regenerate the solvent (amine) and separate the acid gases, heat must be supplied at the bottom of the regeneration column (stripper).

The objective of this project is to design a Reboiler-type heat exchanger that vaporizes the process fluid (consisting of 50% water, 25% ammonia, and 25% benzene by weight) using low-pressure steam (LP-Steam).

## 2 Basis for Selection of Design Parameters (Part A)

In this section, the geometric and operational parameters of the exchanger chosen by the designer are described, along with their technical justification based on engineering standards (TEMA and API) and operating conditions.

### 2.1 Fluid Allocation

Selecting the flow path of each fluid (shell side or tube side) is one of the most critical design steps. In this project:

- **Tube Side: Process Fluid**

- *Reason 1 (Corrosion):* The process fluid contains 25% ammonia, which exhibits strong corrosiveness. Controlling corrosion inside the tubes by selecting an appropriate resistant tube metallurgy is far more economical than fabricating the entire pressure shell from expensive corrosion-resistant alloys.

- *Reason 2 (Pressure):* The operating pressure of the process fluid (7.0 bar) is higher than that of the heating steam (3.5 bar). Standard practice dictates that higher-pressure fluids should flow inside the tubes to minimize the required shell wall thickness and vessel fabrication cost.

- **Shell Side: LP-Steam**

- *Reason:* Steam is a clean, non-corrosive fluid with low viscosity. Condensation of steam across the external surfaces of the bundle provides an extremely high convective heat transfer coefficient and exhibits a minimal fouling factor (approximately  $0.00018 \text{ m}^2\text{K/W}$ ).

## 2.2 Exchanger Type Selection

Based on the TEMA standard, a **BEM**-type heat exchanger was selected:

- **Front Head – Type B (Bonnet):** The simplest and most economical header type. Since frequent internal inspection of the tubes is not necessary for this service, the bolted bonnet provides an optimal and secure closure.
- **Shell – Type E (One Pass):** The standard single-pass shell configuration for phase-change and single-phase industrial thermal duties.
- **Rear Head – Type M (Fixed Tubesheet):** Since the shellside fluid (condensing steam) is exceptionally clean with minimal fouling tendency, there is no operational need for bundle pullout for mechanical cleaning (chemical cleaning is fully sufficient). Fixed-tubesheet exchangers feature the lowest capital cost, and due to the complete absence of internal floating-head gaskets, internal leakage risks of toxic ammonia and volatile benzene are eliminated, ensuring safe high-pressure containment.

## 2.3 Material Selection

- **Tube Material:** Austenitic Stainless Steel 304 (18Cr, 8Ni)

*Justification:* The presence of ammonia in the process fluid strictly rules out the use of copper-based alloys (such as brass or bronze), as ammonia aggressively attacks copper alloys via stress corrosion and dissolution. Stainless Steel 304 provides proven corrosion immunity in hot ammonia-water-benzene environments.

- **Shell Material:** Carbon Steel

*Justification:* The shell envelope contacts only utility steam, which is non-corrosive under normal water treatment. Carbon steel provides maximum structural strength at minimal economic cost.

## 2.4 Tube Geometric Specifications and Layout

- **Tube Outer Diameter and Thickness:** OD = 22.225 mm (7/8 inch), average wall thickness  $t_w = 2.108$  mm. This geometry represents an established refinery standard that provides an ideal trade-off between heat transfer area, mechanical rigidity, and hydraulic pressure drop.
- **Tube Length:** 6.706 m. Sizing with longer tubes reduces the shell internal diameter, resulting in a slimmer, longer vessel that significantly reduces fabrication and flange costs compared to large-diameter shells.
- **Tube Layout Pattern:** 30° Triangular pitch.

*Justification:* A 30° triangular pitch provides the maximum packing density and highest heat transfer area per unit volume. Because mechanical lane cleaning on the shell side is not required for condensing steam, a square layout (which introduces excessive dead space) is unnecessary.

- **Pitch Ratio:** 1.33 (Pitch = 29.559 mm). TEMA requires a minimum center-to-center pitch ratio of 1.25. A ratio of 1.33 ensures adequate ligament thickness for reliable mechanical tube expansion into the tubesheet.

## 2.5 Baffle Geometry

- **Type:** Single-Segmental Carbon Steel baffles
- **Baffle Cut:** 25% diametral cut
- **Spacing:** Central spacing of approximately 233 mm (24 crosspasses)

This layout was selected to promote crossflow across the condensing vapor while providing structural support to eliminate flow-induced acoustic and mechanical vibrations.

## 3 Summary of Design Operating Conditions

Table 1 summarizes the thermodynamic and operational boundary conditions applied in HTRI based on project criteria and engineering decisions.

| Parameter                                                     | Tube Side (Process Fluid)        | Shell Side (LP-Steam)              |
|---------------------------------------------------------------|----------------------------------|------------------------------------|
| Mass Flow Rate                                                | 25,000 kg/hr                     | 5646.8 kg/hr (Calculated)          |
| Temperature (In $\rightarrow$ Out)                            | 70.00°C $\rightarrow$ 111.33°C   | 138.94°C $\rightarrow$ 137.61°C    |
| Operating Pressure                                            | 7.0 bar (700 kPa)                | 3.5 bar (350 kPa)                  |
| Vapor Fraction ( $x_{\text{in}} \rightarrow x_{\text{out}}$ ) | 0.0 $\rightarrow$ 0.50 (Boiling) | 1.0 $\rightarrow$ 0.0 (Condensing) |
| Fouling Factor                                                | 0.000350 m <sup>2</sup> K/W      | 0.000180 m <sup>2</sup> K/W        |
| Design Pressure                                               | 22 bar (2200 kPaG)               | 10 bar (1000 kPaG)                 |
| Design Temperature                                            | 200°C                            | 150°C                              |
| Number of Tube Passes                                         | 2 passes                         | 1 pass                             |

Table 1: Summary of process boundary conditions and operational inputs.

**Note on the Number of Tube Passes:** Although a single tube pass was initially assumed as a baseline, the final design incorporates 2 tube passes. This configuration increases tubeside fluid velocity, enhances the boiling heat transfer coefficient, and mitigates stratified flow and localized film boiling risks.

### 3.1 Conclusion of Part A

Based on the parameters above, a TEMA BEM heat exchanger with 285 Stainless Steel 304 tubes of 6.706 m length was designed. This configuration delivers the required process duty of 3.41 MW while satisfying all hydraulic pressure drop and fouling constraints.

### 3.2 HTRI Specification Sheet – Base Case (25,000 kg/hr)

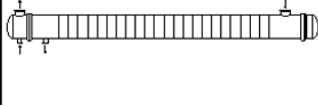
| HTRI                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     | HEAT EXCHANGER SPECIFICATION SHEET |                      |                                           |              | Page 1<br>SI Units |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
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| Customer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                     |                                    |                      | Job No.                                   |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Address                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |                                    |                      | Reference No.                             |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Plant Location                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                     |                                    |                      | Proposal No.                              |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Service of Unit                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                    |                      | Date 2/7/2026 Rev                         |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Size 584.2 x 6706 mm Type BEM Horizontal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                     |                                    |                      | Item No.                                  |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Surf/Unit (Gross/Eff) 133.44 / 131.04 m2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                     |                                    |                      | Connected In 1 Parallel 1 Series          |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Shell/Unit 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                     |                                    |                      | Surf/Shell (Gross/Eff) 133.44 / 131.04 m2 |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| PERFORMANCE OF ONE UNIT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |                                    |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Fluid Allocation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                     | Shell Side                         |                      | Tube Side                                 |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Fluid Name                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     | LP-Steam                           |                      | Process                                   |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Fluid Quantity, Total kg/hr                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                     | 5646.8                             |                      | 25000                                     |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Vapor (In/Out)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                     | 5646.8                             |                      | 12500                                     |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Liquid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                     | 5646.8                             |                      | 12500                                     |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Steam                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                     | 5646.8                             |                      | 1127.0                                    |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Water                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                     | 5646.8                             |                      | 5740.0                                    |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Noncondensables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                    |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Temperature (In/Out) C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                     | 138.94                             |                      | 70.00                                     |              | 111.33             |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Specific Gravity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                     | 0.9155                             |                      | 0.8183                                    |              | 0.8229             |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Viscosity mN-s/m2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     | 0.0135                             |                      | 0.3007                                    |              | 0.0110 V/L 0.2094  |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Molecular Weight, Vapor                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                     |                                    |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Molecular Weight, Noncondensables                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |                                    |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Specific Heat kJ/kg-C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                     | 1.9544                             |                      | 2.9816                                    |              | 1.6592 V/L 3.3932  |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Thermal Conductivity W/m-C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                     | 0.0276                             |                      | 0.3080                                    |              | 0.0263 V/L 0.3803  |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Latent Heat kJ/kg                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     | 2164.5                             |                      | 1008.1                                    |              | 653.58             |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Inlet Pressure kPa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                     | 350.00                             |                      | 700.00                                    |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Velocity m/s                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                     | 8.25                               |                      | 5.01                                      |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Pressure Drop, Allow/Calc kPa                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     | 30.000                             |                      | 12.974                                    |              | 30.000 19.999      |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Fouling Resistance (min) m2-K/W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     | 0.000180                           |                      | 0.000350                                  |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Heat Exchanged 3406885 W                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                     | MTD (Corrected) 38.0 C             |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Transfer Rate, Service 683.40 W/m2-K                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     | Clean 1623.7 W/m2-K                |                      | Actual 814.46 W/m2-K                      |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| CONSTRUCTION OF ONE SHELL                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                     |                                    |                      | Sketch (Bundle/Nozzle Orientation)        |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Design/Test Pressure kPaG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                     | 1000.0 /                           |                      | 2200.0 /                                  |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Design Temperature C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     | 150.00                             |                      | 200.00                                    |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| No Passes per Shell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | 1                                  |                      | 2                                         |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Corrosion Allowance mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                     | 3.200                              |                      | 3.200                                     |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Connections In mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     | 1 @ 205.00                         |                      | 1 @ 77.927                                |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Size & Out mm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     | 1 @ 102.26                         |                      | 1 @ 205.00                                |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Rating Intermediate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     | @                                  |                      | @                                         |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     |                                    |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| <table border="1"> <tr> <td>Tube No.</td> <td>285</td> <td>OD</td> <td>22.225 mm</td> <td>Thk(Avg)</td> <td>2.108 mm</td> <td>Length</td> <td>6.706 m</td> <td>Pitch</td> <td>29.559 mm</td> </tr> <tr> <td>Tube Type</td> <td>Plain</td> <td></td> <td></td> <td></td> <td></td> <td>Material</td> <td>304 Stainless steel (18 Cr, 8 Ni)</td> <td>Tube pattern</td> <td>30</td> </tr> <tr> <td>Shell</td> <td>Carbon steel</td> <td>ID</td> <td>584.20</td> <td>OD</td> <td>603.25 mm</td> <td></td> <td>Shell Cover</td> <td></td> <td></td> </tr> <tr> <td>Channel or Bonnet</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>Channel Cover</td> <td></td> <td></td> </tr> <tr> <td>Tubesheet-Stationary</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>Tubesheet-Floating</td> <td></td> <td></td> </tr> <tr> <td>Floating Head Cover</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>Impingement Plate</td> <td>Rods</td> <td></td> </tr> <tr> <td>Baffles-Cross</td> <td>Type</td> <td>Single-Seg.</td> <td>%Cut (Diam)</td> <td>25</td> <td>Spacing(c/c)</td> <td>232.97</td> <td>Inlet</td> <td>786.11 mm</td> <td></td> </tr> <tr> <td>Baffles-Long</td> <td></td> <td></td> <td>Seal Type</td> <td>None</td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Supports-Tube</td> <td></td> <td></td> <td>U-Bend</td> <td></td> <td>Type</td> <td>None</td> <td></td> <td></td> <td></td> </tr> <tr> <td>Bypass Seal Arrangement</td> <td>0</td> <td>pairs seal strips</td> <td>Tube-Tubesheet Joint</td> <td>Expanded (No groove)</td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Expansion Joint</td> <td></td> <td></td> <td>Type</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Rho-V2-Inlet Nozzle</td> <td>1199.8</td> <td>kg/m-s2</td> <td>Bundle Entrance</td> <td>137.52</td> <td>Bundle Exit</td> <td>2.11</td> <td>kg/m-s2</td> <td></td> <td></td> </tr> <tr> <td>Gaskets-Shell Side</td> <td>Mach. Mtl. (Kammprofile\Flex. Face)</td> <td></td> <td>Tube Side</td> <td>Mach. Mtl. (Kammprofile\Flex. Face)</td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>- Floating Head</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> <td></td> </tr> <tr> <td>Code Requirements</td> <td></td> <td></td> <td></td> <td></td> <td></td> <td>TEMA Class</td> <td>R</td> <td></td> <td></td> </tr> <tr> <td>Weight/Shell</td> <td>3913.1</td> <td>kg</td> <td>Filled with Water</td> <td>5755.8</td> <td>kg</td> <td>Bundle</td> <td>2328.3</td> <td>kg</td> <td></td> </tr> <tr> <td colspan="10">Remarks:</td> </tr> </table> |                                     |                                    |                      |                                           |              |                    |                                   | Tube No.     | 285       | OD | 22.225 mm | Thk(Avg) | 2.108 mm | Length | 6.706 m | Pitch | 29.559 mm | Tube Type | Plain |  |  |  |  | Material | 304 Stainless steel (18 Cr, 8 Ni) | Tube pattern | 30 | Shell | Carbon steel | ID | 584.20 | OD | 603.25 mm |  | Shell Cover |  |  | Channel or Bonnet |  |  |  |  |  |  | Channel Cover |  |  | Tubesheet-Stationary |  |  |  |  |  |  | Tubesheet-Floating |  |  | Floating Head Cover |  |  |  |  |  |  | Impingement Plate | Rods |  | Baffles-Cross | Type | Single-Seg. | %Cut (Diam) | 25 | Spacing(c/c) | 232.97 | Inlet | 786.11 mm |  | Baffles-Long |  |  | Seal Type | None |  |  |  |  |  | Supports-Tube |  |  | U-Bend |  | Type | None |  |  |  | Bypass Seal Arrangement | 0 | pairs seal strips | Tube-Tubesheet Joint | Expanded (No groove) |  |  |  |  |  | Expansion Joint |  |  | Type |  |  |  |  |  |  | Rho-V2-Inlet Nozzle | 1199.8 | kg/m-s2 | Bundle Entrance | 137.52 | Bundle Exit | 2.11 | kg/m-s2 |  |  | Gaskets-Shell Side | Mach. Mtl. (Kammprofile\Flex. Face) |  | Tube Side | Mach. Mtl. (Kammprofile\Flex. Face) |  |  |  |  |  | - Floating Head |  |  |  |  |  |  |  |  |  | Code Requirements |  |  |  |  |  | TEMA Class | R |  |  | Weight/Shell | 3913.1 | kg | Filled with Water | 5755.8 | kg | Bundle | 2328.3 | kg |  | Remarks: |  |  |  |  |  |  |  |  |  |
| Tube No.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 285                                 | OD                                 | 22.225 mm            | Thk(Avg)                                  | 2.108 mm     | Length             | 6.706 m                           | Pitch        | 29.559 mm |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Tube Type                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Plain                               |                                    |                      |                                           |              | Material           | 304 Stainless steel (18 Cr, 8 Ni) | Tube pattern | 30        |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Shell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Carbon steel                        | ID                                 | 584.20               | OD                                        | 603.25 mm    |                    | Shell Cover                       |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Channel or Bonnet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |                                    |                      |                                           |              |                    | Channel Cover                     |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Tubesheet-Stationary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                     |                                    |                      |                                           |              |                    | Tubesheet-Floating                |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Floating Head Cover                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                     |                                    |                      |                                           |              |                    | Impingement Plate                 | Rods         |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Baffles-Cross                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Type                                | Single-Seg.                        | %Cut (Diam)          | 25                                        | Spacing(c/c) | 232.97             | Inlet                             | 786.11 mm    |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Baffles-Long                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                     |                                    | Seal Type            | None                                      |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Supports-Tube                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                     |                                    | U-Bend               |                                           | Type         | None               |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Bypass Seal Arrangement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 0                                   | pairs seal strips                  | Tube-Tubesheet Joint | Expanded (No groove)                      |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Expansion Joint                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                    | Type                 |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Rho-V2-Inlet Nozzle                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1199.8                              | kg/m-s2                            | Bundle Entrance      | 137.52                                    | Bundle Exit  | 2.11               | kg/m-s2                           |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Gaskets-Shell Side                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Mach. Mtl. (Kammprofile\Flex. Face) |                                    | Tube Side            | Mach. Mtl. (Kammprofile\Flex. Face)       |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| - Floating Head                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                     |                                    |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Code Requirements                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                     |                                    |                      |                                           |              | TEMA Class         | R                                 |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Weight/Shell                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 3913.1                              | kg                                 | Filled with Water    | 5755.8                                    | kg           | Bundle             | 2328.3                            | kg           |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
| Remarks:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                     |                                    |                      |                                           |              |                    |                                   |              |           |    |           |          |          |        |         |       |           |           |       |  |  |  |  |          |                                   |              |    |       |              |    |        |    |           |  |             |  |  |                   |  |  |  |  |  |  |               |  |  |                      |  |  |  |  |  |  |                    |  |  |                     |  |  |  |  |  |  |                   |      |  |               |      |             |             |    |              |        |       |           |  |              |  |  |           |      |  |  |  |  |  |               |  |  |        |  |      |      |  |  |  |                         |   |                   |                      |                      |  |  |  |  |  |                 |  |  |      |  |  |  |  |  |  |                     |        |         |                 |        |             |      |         |  |  |                    |                                     |  |           |                                     |  |  |  |  |  |                 |  |  |  |  |  |  |  |  |  |                   |  |  |  |  |  |            |   |  |  |              |        |    |                   |        |    |        |        |    |  |          |  |  |  |  |  |  |  |  |  |
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Figure 1: HTRI Heat Exchanger Specification Sheet – Base Case (25,000 kg/hr).

### 3.3 HTRI Output Summary – Base Case

Design: Horizontal Multipass Flow TEMA BEM Shell With Single-Segmental Baffles. No Data Check Messages (see Runtime Message Report for Warning Messages).

| Process Conditions                       | Hot Shellside (LP-Steam) | Cold Tubeside (Process) |
|------------------------------------------|--------------------------|-------------------------|
| Flow Rate (kg/s)                         | 1.5686                   | 6.9444                  |
| Inlet/Outlet Vapor Fraction (Wt.)        | 1.0000 / 0.0000          | 0.0000 / 0.5000         |
| Inlet/Outlet Temperature (°C)            | 138.94 / 137.61          | 70.00 / 111.33          |
| Inlet Pressure / Avg (kPa)               | 350.00 / 343.51          | 700.00 / 690.00         |
| $\Delta P$ Calculated / Allowable (kPa)  | 12.974 / 30.000          | 19.999 / 30.000         |
| Fouling Resistance (m <sup>2</sup> -K/W) | 0.000180                 | 0.000350                |

| Exchanger Performance           |            |                                    |        |
|---------------------------------|------------|------------------------------------|--------|
| Shell $h$ (W/m <sup>2</sup> -K) | 16,090     | Actual $U$ (W/m <sup>2</sup> -K)   | 814.46 |
| Tube $h$ (W/m <sup>2</sup> -K)  | 2971.9     | Required $U$ (W/m <sup>2</sup> -K) | 683.40 |
| Hot Regime                      | Transition | Duty (MW)                          | 3.4069 |
| Cold Regime                     | Transition | Effective Area (m <sup>2</sup> )   | 131.04 |
| EMTD (°C)                       | 38.0       | Overdesign (%)                     | 19.18  |

| Shell Geometry    |        | Baffle Geometry      |                  |
|-------------------|--------|----------------------|------------------|
| TEMA Type         | BEM    | Baffle Type          | Single-Segmental |
| Shell ID (mm)     | 584.20 | Baffle Cut (% Dia.)  | 25               |
| Series / Parallel | 1 / 1  | Central Spacing (mm) | 232.97           |
| Orientation (deg) | 0.00   | Crosspasses          | 24               |

| Tube Geometry        |              | Nozzles                       |                  |
|----------------------|--------------|-------------------------------|------------------|
| Tube Type            | Plain        | Shell Inlet / Outlet (mm)     | 205.00 / 102.26  |
| Tube OD (mm)         | 22.225       | Shell Nozzle Heights (In/Out) | 28.36 / 15.13 mm |
| Tube Length (m)      | 6.706        | Tube Inlet / Outlet (mm)      | 77.93 / 205.00   |
| Pitch Ratio / Layout | 1.3300 / 30° | Tube Pass Count               | 2                |
| Tube Count           | 285          |                               |                  |

| Thermal Resistance Breakdown (%) |       | Velocities (m/s) |       |       |
|----------------------------------|-------|------------------|-------|-------|
| Shell Condensing Film            | 5.06  |                  | Min   | Max   |
| Tube Boiling Film                | 33.82 | Tubeside         | 0.24  | 10.50 |
| Fouling Resistance               | 49.84 | Crossflow        | 0.19  | 13.74 |
| Tube Metal Wall                  | 11.28 | Window           | 0.026 | 13.75 |
| Flow Fractions                   |       |                  |       |       |
| Stream A                         | 0.198 |                  |       |       |
| Stream B                         | 0.470 |                  |       |       |
| Stream C                         | 0.100 |                  |       |       |
| Stream E                         | 0.149 |                  |       |       |
| Stream F                         | 0.083 |                  |       |       |

Table 2: HTRI Output Summary – Base Case (25,000 kg/hr).

## 4 Performance Analysis Under Increased Capacity (Part B)

### 4.1 Comparison of Design Data

Table 3 presents a side-by-side comparison between the baseline design (25,000 kg/hr) and the debottlenecked case (30,000 kg/hr). In this analysis, tube diameter and tube length were held constant to isolate the net impact of the throughput increase.

| Parameter                    | Base Design              | Expanded Design           | % Change / Engineering Notes           |
|------------------------------|--------------------------|---------------------------|----------------------------------------|
| Tubeside Mass Flow Rate      | 25,000 kg/hr             | 30,000 kg/hr              | +20.0%                                 |
| Thermal Duty ( $Q$ )         | 3.407 MW                 | 4.086 MW                  | +19.9% (Directly proportional)         |
| Tube Outer Diameter          | 22.225 mm                | 22.225 mm                 | Fixed (Standard)                       |
| Tubeside Passes              | 2 passes                 | 2 passes                  | Fixed (Maintains velocity)             |
| Total Tube Count             | 285                      | 333                       | Increased to provide required area     |
| Tubeside Pressure Drop       | 19.99 kPa                | 21.56 kPa                 | Modest increase (< 30 kPa allowable)   |
| Overall Heat Transfer Coeff. | 814.5 W/m <sup>2</sup> K | 823.49 W/m <sup>2</sup> K | +1.1% (Enhanced convective turbulence) |
| Overdesign Margin            | 19.18%                   | 17.86%                    | Maintained at safe operational levels  |

Table 3: Comparison of thermal-hydraulic data between base and increased-capacity operations.

### 4.2 Technical Analysis of Results

#### 4.2.1 Thermal Capability at High Flow Rate

Software convergence indicates that when the flow rate is scaled to 30,000 kg/hr:

- **Thermal Duty:** Expands to 4.086 MW, achieving the target outlet vapor fraction ( $x = 0.50$ ).

- **Overdesign Margin:** Remains robust at 17.86%. This confirms that the unit operates reliably under expanded capacity while retaining adequate surplus area to absorb long-term fouling.

#### 4.2.2 Hydraulic Behavior and Pressure Losses

- **Tubeside Pressure Drop:** Evaluated at 21.56 kPa, well below the allowable limit of 30.0 kPa.
- **Fluid Velocity:** The in-tube velocity (5.10 m/s max mixture velocity) remains within allowable erosion and vibration thresholds, maintaining a high convective film coefficient.

## 4.3 HTRI Specification Sheet – Increased-Capacity Case (30,000 kg/hr)

| HTRI                                                                                                 |  | HEAT EXCHANGER SPECIFICATION SHEET |  | Page 1<br>SI Units     |  |
|------------------------------------------------------------------------------------------------------|--|------------------------------------|--|------------------------|--|
| Customer                                                                                             |  | Job No.                            |  |                        |  |
| Address                                                                                              |  | Reference No.                      |  |                        |  |
| Plant Location                                                                                       |  | Proposal No.                       |  |                        |  |
| Service of Unit                                                                                      |  | Date                               |  | 2/7/2026 Rev           |  |
| Size                                                                                                 |  | Item No.                           |  |                        |  |
| Surf/Unit (Gross/Eff)                                                                                |  | Connected In                       |  | 1 Parallel 1 Series    |  |
| 635 x 6706 mm Type BEM Horizontal                                                                    |  | Surf/Shell (Gross/Eff)             |  | 155.92 / 152.97 m2     |  |
| PERFORMANCE OF ONE UNIT                                                                              |  |                                    |  |                        |  |
| Fluid Allocation                                                                                     |  | Shell Side                         |  | Tube Side              |  |
| Fluid Name                                                                                           |  | LP-Steam                           |  | Process                |  |
| Fluid Quantity, Total kg/hr                                                                          |  | 6775.0                             |  | 30000                  |  |
| Vapor (In/Out)                                                                                       |  | 6775.0                             |  | 15000                  |  |
| Liquid                                                                                               |  | 6775.0                             |  | 15000                  |  |
| Steam                                                                                                |  | 6775.0                             |  | 1351.6                 |  |
| Water                                                                                                |  | 6775.0                             |  | 6888.4                 |  |
| Noncondensables                                                                                      |  |                                    |  |                        |  |
| Temperature (In/Out) C                                                                               |  | 138.94                             |  | 70.00                  |  |
| Specific Gravity                                                                                     |  | 0.9154                             |  | 0.8183                 |  |
| Viscosity mN-s/m2                                                                                    |  | 0.0135                             |  | 0.0110 V/L 0.2096      |  |
| Molecular Weight, Vapor                                                                              |  |                                    |  |                        |  |
| Molecular Weight, Noncondensables                                                                    |  |                                    |  |                        |  |
| Specific Heat kJ/kg-C                                                                                |  | 1.9544                             |  | 2.9816                 |  |
| Thermal Conductivity W/m-C                                                                           |  | 0.0276                             |  | 0.0262 V/L 0.3803      |  |
| Latent Heat kJ/kg                                                                                    |  | 2164.5                             |  | 1008.1                 |  |
| Inlet Pressure kPa                                                                                   |  | 350.00                             |  | 700.00                 |  |
| Velocity m/s                                                                                         |  | 8.37                               |  | 5.10                   |  |
| Pressure Drop, Allow/Calc kPa                                                                        |  | 30.000                             |  | 30.000                 |  |
| Fouling Resistance (min) m2-K/W                                                                      |  | 0.000180                           |  | 0.000350               |  |
| Heat Exchanged                                                                                       |  | 4086328 W                          |  | MTD (Corrected) 38.2 C |  |
| Transfer Rate, Service                                                                               |  | 698.71 W/m2-K                      |  | Actual 823.49 W/m2-K   |  |
| CONSTRUCTION OF ONE SHELL                                                                            |  |                                    |  |                        |  |
|                                                                                                      |  | Shell Side                         |  | Tube Side              |  |
| Design/Test Pressure kPaG                                                                            |  | 1000.0 /                           |  | 2200.0 /               |  |
| Design Temperature C                                                                                 |  | 150.00                             |  | 200.00                 |  |
| No Passes per Shell                                                                                  |  | 1                                  |  | 2                      |  |
| Corrosion Allowance mm                                                                               |  | 3.200                              |  | 3.200                  |  |
| Connections                                                                                          |  | In mm                              |  | 1 @ 258.88             |  |
| Size & Rating                                                                                        |  | Out mm                             |  | 1 @ 77.927             |  |
|                                                                                                      |  | Intermediate                       |  | 1 @ 258.88             |  |
|                                                                                                      |  | @                                  |  | @                      |  |
| Sketch (Bundle/Nozzle Orientation)                                                                   |  |                                    |  |                        |  |
|                  |  |                                    |  |                        |  |
| Tube No. 333 OD 22.225 mm Thk(Avg) 2.108 mm Length 6.706 m Pitch 29.559 mm                           |  |                                    |  |                        |  |
| Tube Type Plain Material 304 Stainless steel (18 Cr, 8 Ni) Tube pattern 30                           |  |                                    |  |                        |  |
| Shell Carbon steel ID 635.00 OD 654.05 mm Shell Cover                                                |  |                                    |  |                        |  |
| Channel or Bonnet Channel Cover                                                                      |  |                                    |  |                        |  |
| Tubesheet-Stationary Tubesheet-Floating                                                              |  |                                    |  |                        |  |
| Floating Head Cover Impingement Plate Rods                                                           |  |                                    |  |                        |  |
| Baffles-Cross Type Single-Seg. %Cut (Diam) 25 Spacing(c/c) 264.43 Inlet 837.25 mm                    |  |                                    |  |                        |  |
| Baffles-Long Seal Type None                                                                          |  |                                    |  |                        |  |
| Supports-Tube U-Bend Type None                                                                       |  |                                    |  |                        |  |
| Bypass Seal Arrangement 0 pairs seal strips Tube-Tubesheet Joint Expanded (No groove)                |  |                                    |  |                        |  |
| Expansion Joint Type                                                                                 |  |                                    |  |                        |  |
| Rho-V2-Inlet Nozzle 679.18 kg/m-s2 Bundle Entrance 142.22 Bundle Exit 1.66 kg/m-s2                   |  |                                    |  |                        |  |
| Gaskets-Shell Side Mach. Mtl. (Kammprofile\Flex. Face) Tube Side Mach. Mtl. (Kammprofile\Flex. Face) |  |                                    |  |                        |  |
| - Floating Head                                                                                      |  |                                    |  |                        |  |
| Code Requirements TEMA Class R                                                                       |  |                                    |  |                        |  |
| Weight/Shell 4451.3 kg Filled with Water 6698.1 kg Bundle 2713.4 kg                                  |  |                                    |  |                        |  |
| Remarks:                                                                                             |  |                                    |  |                        |  |
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Figure 2: HTRI Heat Exchanger Specification Sheet – Increased-Capacity Case (30,000 kg/hr).

## 4.4 HTRI Output Summary – Increased-Capacity Case

Design: Horizontal Multipass Flow TEMA BEM Shell With Single-Segmental Baffles. No Data Check Messages (see Runtime Message Report for Warning Messages).

| Process Conditions                       | Hot Shellside (LP-Steam) | Cold Tubeside (Process) |
|------------------------------------------|--------------------------|-------------------------|
| Flow Rate (kg/s)                         | 1.8819                   | 8.3333                  |
| Inlet/Outlet Vapor Fraction (Wt.)        | 1.0000 / 0.0000          | 0.0000 / 0.5000         |
| Inlet/Outlet Temperature (°C)            | 138.94 / 137.75          | 70.00 / 111.25          |
| Inlet Pressure / Avg (kPa)               | 350.00 / 344.21          | 700.00 / 689.22         |
| $\Delta P$ Calculated / Allowable (kPa)  | 11.572 / 30.000          | 21.561 / 30.000         |
| Fouling Resistance (m <sup>2</sup> -K/W) | 0.000180                 | 0.000350                |

| Exchanger Performance           |            |                                    |        |
|---------------------------------|------------|------------------------------------|--------|
| Shell $h$ (W/m <sup>2</sup> -K) | 15,854     | Actual $U$ (W/m <sup>2</sup> -K)   | 823.49 |
| Tube $h$ (W/m <sup>2</sup> -K)  | 3078.6     | Required $U$ (W/m <sup>2</sup> -K) | 698.71 |
| Hot Regime                      | Transition | Duty (MW)                          | 4.0863 |
| Cold Regime                     | Transition | Effective Area (m <sup>2</sup> )   | 152.97 |
| EMTD (°C)                       | 38.2       | Overdesign (%)                     | 17.86  |

| Shell Geometry    |        | Baffle Geometry      |                  |
|-------------------|--------|----------------------|------------------|
| TEMA Type         | BEM    | Baffle Type          | Single-Segmental |
| Shell ID (mm)     | 635.00 | Baffle Cut (% Dia.)  | 25               |
| Series / Parallel | 1 / 1  | Central Spacing (mm) | 264.43           |
| Orientation (deg) | 0.00   | Crosspasses          | 21               |

| Tube Geometry        |              | Nozzles                       |                  |
|----------------------|--------------|-------------------------------|------------------|
| Tube Type            | Plain        | Shell Inlet / Outlet (mm)     | 258.88 / 102.26  |
| Tube OD (mm)         | 22.225       | Shell Nozzle Heights (In/Out) | 40.30 / 21.25 mm |
| Tube Length (m)      | 6.706        | Tube Inlet / Outlet (mm)      | 77.93 / 258.88   |
| Pitch Ratio / Layout | 1.3300 / 30° | Tube Pass Count               | 2                |
| Tube Count           | 333          |                               |                  |

| Thermal Resistance Breakdown (%) |       | Velocities (m/s) |       |       |
|----------------------------------|-------|------------------|-------|-------|
| Shell Condensing Film            | 5.19  |                  | Min   | Max   |
| Tube Boiling Film                | 33.01 | Tubeside         | 0.24  | 10.94 |
| Fouling Resistance               | 50.39 | Crossflow        | 0.20  | 13.48 |
| Tube Metal Wall                  | 11.40 | Window           | 0.027 | 13.71 |
| Flow Fractions                   |       |                  |       |       |
| Stream A                         | 0.198 |                  |       |       |
| Stream B                         | 0.470 |                  |       |       |
| Stream C                         | 0.106 |                  |       |       |
| Stream E                         | 0.139 |                  |       |       |
| Stream F                         | 0.087 |                  |       |       |

Table 4: HTRI Output Summary – Increased-Capacity Case (30,000 kg/hr).

## 5 Response to the Project Question: Linearity of Area vs. Flow Rate

**Question:** Is the relationship between the percentage increase in process flow rate and the required heat transfer surface area strictly linear (+20% flow  $\rightarrow$  +20% area)?

**Analysis and Answer:** No, the relationship is sub-linear. The required heat transfer area increases by only +16.74% (131.04 m<sup>2</sup>  $\rightarrow$  152.97 m<sup>2</sup>) for a +20.0% expansion in throughput.

Governing formulation:

$$Q = U \cdot A \cdot \Delta T_{LM} \quad (1)$$

1. Increasing process flow rate by 20% proportionally increases the thermal duty  $Q$  by approximately 20%.
2. Simultaneously, higher mass flux elevates fluid velocity inside the tubes, which increases the Reynolds number ( $Re$ ).
3. Because the convective boiling coefficient obeys  $h_i \propto Re^{0.8}$ , higher turbulence increases  $h_{\text{tube}}$  from 2971.9 to 3078.6 W/m<sup>2</sup>K, which in turn raises the overall heat transfer coefficient  $U$  from 814.5 to 823.5 W/m<sup>2</sup>K.
4. Because the overall heat transfer coefficient  $U$  increases, the required heat transfer surface area  $A$  does not need to scale by the full 20%.

**Conclusion:** Capacity expansion improves the convective transport coefficient, causing the required area growth slope to be lower than the mass throughput increase ( $dA/d\dot{m} < A_0/\dot{m}_0$ ).

## 6 Review of Error and Warning Messages (Part C)

*“The calculated or specified terminal temperatures are impossible. Check the specified physical properties and process conditions, and rerun the case. Run Failed.”*

**Resolution of Pressure and Temperature Inconsistency:** The software encountered terminal temperature contradictions during initial thermodynamic calculations. Specifying operating pressures consistently in Pascal units and re-verifying stream thermodynamic property curves eliminated this fatal error.

*“Wavy stratified flow and partial dryout are expected in horizontal tubes. We recommend increasing flow rate or reducing tube diameter to avoid stratified flow because of its poor heat transfer performance and high uncertainty in predicting boiling heat transfer coefficients in this flow regime.”*

**Resolution of Stratified Flow and Dryout Warning:** Selecting a 2-pass tube arrangement increased in-tube liquid-vapor velocity, maintaining an annular/intermittent flow regime and avoiding poor wetted-perimeter heat transfer associated with wavy stratified flow.

*“Impingement rod width ratio has been reduced to fit the array into the shell while permitting a one-half diameter bundle of tubes.”*

**Correction by Adjusting Impingement Protection:** To mitigate shellside inlet erosive forces ( $\rho V^2 > 1000 \text{ kg/m} \cdot \text{s}^2$ ), impingement rods were introduced. The rod diameter ratio was adjusted to fit the array into the shell while preserving bundle entrance area.

*“Unspecified channelside gasket – defaulting to gasket code 5054 – Kammprofile.”*

**Channel Gasket Definition:** Resolved by assigning gasket code 5054 (Kammprofile metal core with flexible graphite facing), which matches high-pressure sour service criteria.

*“ASME Section VIII Div. 1 Code was used to ESTIMATE all pressure vessel dimensions and weights. The dimensions in this report cannot be used to fabricate the vessel.”*

This is a standard informational disclaimer from HTRI stating that mechanical shell thicknesses and vessel weights are preliminary estimates for thermal simulation and must be confirmed via structural code calculations (such as PVElite).

*“Transition boiling has been predicted in at least one increment on the tube side. To confirm, check tubeside monitor for increments in transition boiling.”*

This message indicates localized transition between nucleate and film boiling due to elevated wall heat fluxes. Because the mean temperature driving force is moderate (EMTD  $\approx 38^\circ\text{C}$ ), local wall heat fluxes remain safely below critical heat flux ( $q'' < q''_{\text{CHF}}$ ), ensuring stable operation without sustained dryout.

|                                                                                   |  |                                    |                         |                |             |
|-----------------------------------------------------------------------------------|--|------------------------------------|-------------------------|----------------|-------------|
|  |  | <b>Output Summary</b>              |                         | Page 1         |             |
|                                                                                   |  | Released to the following company: |                         |                |             |
|                                                                                   |  | 989144785264                       |                         |                |             |
| Xist 7.3.2 2/7/2026 22:14 SN: 46639-                                              |  |                                    |                         | SI Units       |             |
| Design - Horizontal Multipass Flow TEMA BEM Shell With Single-Segmental Baffles   |  |                                    |                         |                |             |
| No Data Check Messages.                                                           |  |                                    |                         |                |             |
| See Runtime Message Report for Warning Messages.                                  |  |                                    |                         |                |             |
| Process Conditions                                                                |  | Hot Shellside                      |                         | Cold Tubeside  |             |
| Fluid name                                                                        |  | LP-Steam                           |                         | Process        |             |
| Flow rate (kg/s)                                                                  |  | 1.5686                             |                         | 6.9444         |             |
| Inlet/Outlet Y (Wt. frac vap.)                                                    |  | 1.0000                             | 0.0000                  | 0.0000         | 0.5000      |
| Inlet/Outlet T (Deg C)                                                            |  | 138.94                             | 137.61                  | 70.00          | 111.33      |
| Inlet P/Avg (kPa)                                                                 |  | 350.00                             | 343.51                  | 700.00         | 690.00      |
| dP/Allow. (kPa)                                                                   |  | 12.974                             | 30.000                  | 19.999         | 30.000      |
| Fouling (m2-K/W)                                                                  |  | 0.000180                           |                         | 0.000350       |             |
| Exchanger Performance                                                             |  |                                    |                         |                |             |
| Shell h (W/m2-K)                                                                  |  | 16090                              | Actual U (W/m2-K)       | 814.46         |             |
| Tube h (W/m2-K)                                                                   |  | 2971.9                             | Required U (W/m2-K)     | 683.40         |             |
| Hot regime (--)                                                                   |  | Transition                         | Duty (MegaWatts)        | 3.4069         |             |
| Cold regime (--)                                                                  |  | Tran                               | Eff. area (m2)          | 131.04         |             |
| EMTD (Deg C)                                                                      |  | 38.0                               | Overdesign (%)          | 19.18          |             |
| Shell Geometry                                                                    |  |                                    | Baffle Geometry         |                |             |
| TEMA type (--)                                                                    |  | BEM                                | Baffle type             |                | Single-Seg. |
| Shell ID (mm)                                                                     |  | 584.20                             | Baffle cut (Pct Dia.)   |                | 25          |
| Series (--)                                                                       |  | 1                                  | Baffle orientation (--) |                | Parallel    |
| Parallel (--)                                                                     |  | 1                                  | Central spacing (mm)    |                | 232.97      |
| Orientation (deg)                                                                 |  | 0.00                               | Crosspasses (--)        |                | 24          |
| Tube Geometry                                                                     |  |                                    | Nozzles                 |                |             |
| Tube type (--)                                                                    |  | Plain                              | Shell inlet (mm)        |                | 205.00      |
| Tube OD (mm)                                                                      |  | 22.225                             | Shell outlet (mm)       |                | 102.26      |
| Length (m)                                                                        |  | 6.706                              | Inlet height (mm)       |                | 28.357      |
| Pitch ratio (--)                                                                  |  | 1.3300                             | Outlet height (mm)      |                | 15.127      |
| Layout (deg)                                                                      |  | 30                                 | Tube inlet (mm)         |                | 77.927      |
| Tubecount (--)                                                                    |  | 285                                | Tube outlet (mm)        |                | 205.00      |
| Tube Pass (--)                                                                    |  | 2                                  |                         |                |             |
| Thermal Resistance, %                                                             |  | Velocities, m/s                    |                         | Flow Fractions |             |
| Shell 5.06                                                                        |  | Min Max                            |                         | A 0.198        |             |
| Tube 33.82                                                                        |  | Shellside 0.24                     | 10.50                   | B 0.470        |             |
| Fouling 49.84                                                                     |  | Crossflow 0.19                     | 13.74                   | C 0.100        |             |
| Metal 11.28                                                                       |  | Window 2.64e-2                     | 13.75                   | E 0.149        |             |
|                                                                                   |  |                                    |                         | F 0.083        |             |

Figure 3: HTRI Output.

|                                                                                   |  |                                    |                         |                |             |
|-----------------------------------------------------------------------------------|--|------------------------------------|-------------------------|----------------|-------------|
|  |  | <b>Output Summary</b>              |                         | Page 1         |             |
|                                                                                   |  | Released to the following company: |                         |                |             |
|                                                                                   |  | sharif                             |                         |                |             |
|                                                                                   |  | shayan                             |                         |                |             |
| Xist 7.3.2 2/9/2026 21:51 SN: 46639-                                              |  |                                    |                         | SI Units       |             |
| Design - Horizontal Multipass Flow TEMA BEM Shell With Single-Segmental Baffles   |  |                                    |                         |                |             |
| No Data Check Messages.                                                           |  |                                    |                         |                |             |
| See Runtime Message Report for Warning Messages.                                  |  |                                    |                         |                |             |
| Process Conditions                                                                |  | Hot Shellside                      |                         | Cold Tubeside  |             |
| Fluid name                                                                        |  | LP-Steam                           |                         | Process        |             |
| Flow rate (kg/s)                                                                  |  | 1.8819                             |                         | 8.3333         |             |
| Inlet/Outlet Y (Wt. frac vap.)                                                    |  | 1.0000                             | 0.0000                  | 0.0000         | 0.5000      |
| Inlet/Outlet T (Deg C)                                                            |  | 138.94                             | 137.75                  | 70.00          | 111.25      |
| Inlet P/Avg (kPa)                                                                 |  | 350.00                             | 344.21                  | 700.00         | 689.22      |
| dP/Allow. (kPa)                                                                   |  | 11.572                             | 30.000                  | 21.561         | 30.000      |
| Fouling (m2-K/W)                                                                  |  | 0.000180                           |                         | 0.000350       |             |
| Exchanger Performance                                                             |  |                                    |                         |                |             |
| Shell h (W/m2-K)                                                                  |  | 15854                              | Actual U (W/m2-K)       | 823.49         |             |
| Tube h (W/m2-K)                                                                   |  | 3078.6                             | Required U (W/m2-K)     | 698.71         |             |
| Hot regime (--)                                                                   |  | Transition                         | Duty (MegaWatts)        | 4.0863         |             |
| Cold regime (--)                                                                  |  | Tran                               | Eff. area (m2)          | 152.97         |             |
| EMTD (Deg C)                                                                      |  | 38.2                               | Overdesign (%)          | 17.86          |             |
| Shell Geometry                                                                    |  |                                    | Baffle Geometry         |                |             |
| TEMA type (--)                                                                    |  | BEM                                | Baffle type             |                | Single-Seg. |
| Shell ID (mm)                                                                     |  | 635.00                             | Baffle cut (Pct Dia.)   |                | 25          |
| Series (--)                                                                       |  | 1                                  | Baffle orientation (--) |                | Parallel    |
| Parallel (--)                                                                     |  | 1                                  | Central spacing (mm)    |                | 264.43      |
| Orientation (deg)                                                                 |  | 0.00                               | Crosspasses (--)        |                | 21          |
| Tube Geometry                                                                     |  |                                    | Nozzles                 |                |             |
| Tube type (--)                                                                    |  | Plain                              | Shell inlet (mm)        |                | 258.88      |
| Tube OD (mm)                                                                      |  | 22.225                             | Shell outlet (mm)       |                | 102.26      |
| Length (m)                                                                        |  | 6.706                              | Inlet height (mm)       |                | 40.303      |
| Pitch ratio (--)                                                                  |  | 1.3300                             | Outlet height (mm)      |                | 21.247      |
| Layout (deg)                                                                      |  | 30                                 | Tube inlet (mm)         |                | 77.927      |
| Tubecount (--)                                                                    |  | 333                                | Tube outlet (mm)        |                | 258.88      |
| Tube Pass (--)                                                                    |  | 2                                  |                         |                |             |
| Thermal Resistance, %                                                             |  | Velocities, m/s                    |                         | Flow Fractions |             |
| Shell 5.19                                                                        |  | Min                                | Max                     | A              | 0.198       |
| Tube 33.01                                                                        |  | Tubeside 0.24                      | 10.94                   | B              | 0.470       |
| Fouling 50.39                                                                     |  | Crossflow 0.20                     | 13.48                   | C              | 0.106       |
| Metal 11.40                                                                       |  | Window 2.70e-2                     | 13.71                   | E              | 0.139       |
|                                                                                   |  |                                    |                         | F              | 0.087       |

Figure 4: HTRI Output.

## 7 Summary

The thermal-hydraulic design developed in this project provides a reliable solution across both operating throughputs:

- **Configuration:** TEMA BEM Shell-and-Tube Exchanger ( $D_{\text{shell}} = 584.2$  mm for base design, expanding to 635.0 mm for debottlenecked case).
- **Tubing:** Stainless Steel 304 plain tubes ( $D_o = 22.225$  mm, length  $L = 6.706$  m, 2 passes).
- **Base Operations (25,000 kg/hr):** Requires 285 tubes, transferring 3.41 MW with 19.18% overdesign and a low pressure drop ( $\Delta P_t = 19.99$  kPa,  $\Delta P_s = 12.97$  kPa).
- **Expanded Operations (30,000 kg/hr):** Debottlenecked to 333 tubes, transferring 4.09 MW with 17.86% overdesign and safe hydraulic margins ( $\Delta P_t = 21.56$  kPa < 30 kPa).